Package 'qeshape'

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Type Package

Title Quantile-Expectile Calibration Curves and Symmetry Tests

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Description Estimates quantile-expectile calibration curves from univariate samples using empirical partial deviations. Provides asymmetry curves, simultaneous empirical-bootstrap confidence bands, and centered-bootstrap Cramer-von Mises and supremum tests of distributional symmetry. Exact empirical step-function calculations avoid density estimation, smoothing, and numerical inversion of expectiles.

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`qeshape-package` *Quantile–Expectile Calibration and Symmetry Inference*

Description

Estimate quantile–expectile calibration curves and simultaneous confidence bands. Test distributional symmetry using a centered empirical bootstrap. The implementation uses exact finite grids for empirical suprema and exact integration of the empirical step functions for the unweighted Cramer–von Mises statistic.

Details

Use `qe_curve` for calibration estimates, `qe_asymmetry` for the reflected calibration departure, `qe_band` for simultaneous confidence bands, and `qe_symmetry_test` for symmetry tests. The test result supports the standard `htest` print method; the remaining objects provide `print`, `plot`, `as.data.frame`, and `predict` methods.

Open the installed methods guide with:

```
guide <- system.file("doc", "methods.md", package = "qeshape")
file.show(guide)
```

Statistical scope

For a nondegenerate distribution with a finite first moment, the calibration curve assigns to a quantile the expectile level associated with the same threshold. It is unchanged by translation and positive rescaling.

A nondegenerate distribution with a finite first moment is called *shape regular* when its quantile function is strictly increasing and locally absolutely continuous on $(0, 1)$. On this class, the complete calibration curve identifies distributional shape up to location and positive scale. The reflected population curve characterizes symmetry when it vanishes throughout $(0, 1/2]$.

The inference results assume independent, identically distributed observations, a finite second moment, and a continuous, strictly positive density on a neighborhood of $[Q(e), Q(1-e)]$, where e is the fixed trim and Q is the population quantile function. Critical-value regularity is also required; the individual function topics give the details. Symmetry testing additionally uses shape regularity under the null. Accepting ties or rounded observations computationally does not establish these assumptions.

Tests with fixed trimming detect departures visible on that interval. Nonrejection does not establish global symmetry. Separate bands for two samples are descriptive and do not provide a calibrated two-sample test.

Workflow

Inspect a curve with `qe_curve` and an asymmetry curve with `qe_asymmetry`. Use `qe_band` for simultaneous uncertainty and `qe_symmetry_test` for a formal test. Choose the trim and test statistic before inspecting test results. The default statistic is Cramer–von Mises; the companion supremum statistic can be conservative in finite samples. Increase the number of bootstrap replications for a final analysis. The examples use smaller values to run quickly.

Author(s)

Guanjie Lyu

References

Lyu, G. (2026). *Quantile–expectile calibration curves for distributional shape: estimation, inference, and symmetry testing*. Unpublished manuscript.

See Also

`qe_curve`, `qe_asymmetry`, `qe_band`, `qe_symmetry_test`, `qe_methods`

Examples

```
set.seed(2026)
x <- rlnorm(150)

curve <- qe_curve(x)
predict(curve, probs = c(0.1, 0.5, 0.9))
plot(curve)

asymmetry <- qe_asymmetry(x)
plot(asymmetry)

# Small B for illustration; increase it for a final analysis.
band <- qe_band(x, trim = 0.05, B = 99, seed = 7)
predict(band, probs = c(0.1, 0.5, 0.9))
plot(band)

test <- qe_symmetry_test(x, method = "cvm", trim = 0.05,
                         B = 99, seed = 7)
test
```

---

qe_asymmetry *Estimate the Calibration Asymmetry Curve*

Description

Evaluate the departure of the reflected empirical calibration sum from one. For a shape-regular population distribution, the corresponding curve vanishes under symmetry.

Usage

```
qe_asymmetry(x, probs = (5:50) / 100, na.rm = FALSE)
```

Arguments

x	A nonempty finite real numeric vector of observations.
probs	A nonempty numeric vector with values strictly above zero and at most $1/2$. Order and duplicates are preserved.
na.rm	A logical scalar. If TRUE, remove NA and NaN. Infinite values are always rejected.

Details

The estimated asymmetry curve is

$$\widehat{D}(p) = \widehat{\tau}(p) + \widehat{\tau}(1-p) - 1, \qquad 0 < p \leq 1/2.$$

Calibration uses the empirical left quantile convention described in qe_curve. Reflected ranks are computed on the same probability grid to avoid cancellation when forming $1-p$. A constant sample has asymmetry zero by convention.

Shape regularity is defined in qeshape-package. On this class, a population asymmetry curve that is zero over the entire half interval characterizes symmetry. A curve evaluated at finitely many points or on a trimmed interval does not establish this global property.

Empirical left quantiles have jumps. Even an exactly symmetric finite sample can therefore have nonzero asymmetry at probability breakpoints. The zero-curve characterization concerns the regular population curve.

Value

An object of class `qe_asymmetry`, with components:

probabilities The requested probabilities.

asymmetry Estimated asymmetry values.

n Number of observations used.

omitted Number of missing observations removed.

call The matched function call.

Sample information is retained for exact prediction; internal components are not part of the public interface.

References

Lyu, G. (2026). *Quantile–expectile calibration curves for distributional shape: estimation, inference, and symmetry testing*. Unpublished manuscript, Section 4.

See Also

`qe_curve`, `qe_symmetry_test`, `qe_methods`

Examples

```
x <- c(-2, -1, 0, 1, 4)
fit <- qe_asymmetry(x, probs = c(0.1, 0.25, 0.5))
as.data.frame(fit)
predict(fit, probs = 0.2)

# The empirical left median is the lower observation when n = 2.
qe_asymmetry(c(-1, 1), probs = c(0.25, 0.5))
```

---

qe_band *Construct a Simultaneous Calibration Confidence Band*

Description

Construct an unstudentized, simultaneous empirical-bootstrap confidence band for the calibration curve on a fixed trimmed probability interval.

Usage

```
qe_band(x, trim = 0.05, conf.level = 0.95, B = 999L,
        seed = NULL, na.rm = FALSE)
```

Arguments

x	A finite real numeric vector. After missing-value removal it must contain at least two observations and must be nonconstant.
trim	A finite numeric scalar strictly between zero and $1/2$. The confidence band covers `[trim, 1 - trim]`. Choose this value independently of the data. Values so small that `1 - trim` rounds to one in double precision are rejected.
conf.level	A finite numeric scalar strictly between zero and one.
B	A positive integer number of bootstrap replications.
seed	`NULL` or a nonnegative integer seed. With `NULL`, the caller's random-number stream is used and advanced. A supplied seed is scoped to this call, restoring `.Random.seed` and the RNG kind on exit. A cached normal draw from R's `"Box-Muller"` generator cannot be restored; see Details.
na.rm	A logical scalar. If `TRUE`, remove `NA` and `NaN`. Infinite values are always rejected.

Details

Each bootstrap draw resamples the empirical distribution through multinomial observation counts. The centered process is

$$Z^*(p) = \sqrt{n}\{\widehat{\tau}^*(p) - \widehat{\tau}(p)\}.$$

Its absolute supremum is calculated exactly on the two trim endpoints and all interior k/n. Let c be the `conf.level` empirical quantile of the bootstrap suprema, using `quantile(type = 1)`. The band has untruncated half width $c/\sqrt{n}$; endpoints are clipped to $[0, 1]$. No density or variance estimation is required. Constant bootstrap resamples use the calibration convention $1/2$.

R's `"Box-Muller"` normal generator keeps a cached normal draw outside `.Random.seed`; a scoped seed cannot restore this cache. The default `"Inversion"` normal generator avoids this issue. Use `seed = NULL` when continuation of an existing Box-Muller cache matters. If `.Random.seed` was absent before a seeded call, it is removed on exit. Fixed-seed reproduction also depends on the R random-number settings.

The asymptotic justification assumes iid observations, a finite second moment, and a continuously differentiable distribution with strictly positive density on a neighborhood of $[Q(\mathrm{trim}), Q(1 - \mathrm{trim})]$, where Q is the population quantile function. The distribution of the limiting supremum must be continuous and strictly cross the relevant quantile level. The trim is fixed and deterministic, and Monte Carlo consistency requires B to increase. Ties are supported computationally but do not establish these assumptions. The band is not a finite-sample coverage guarantee.

Value

An object of classes `qe_band` and `qe_curve`, containing the components documented in `qe_curve`, evaluated on the exact supremum grid, and:

lower, upper Clipped confidence-band endpoints.

trim, conf.level, B, seed The supplied inference settings.

critical.value The bootstrap critical value on the $\sqrt{n}$ scale.

half.width The untruncated band half width.

bootstrap.statistics The `B` bootstrap suprema.

References

Lyu, G. (2026). *Quantile–expectile calibration curves for distributional shape: estimation, inference, and symmetry testing*. Unpublished manuscript, Section 3.

See Also

qe_curve, qe_symmetry_test, qe_methods

Examples

```
# Small B for a fast example; increase it for an analysis.
x <- c(-3, -2, -1, 0, 0.5, 1, 2, 4)
fit <- qe_band(x, B = 39, seed = 7)
fit
predict(fit, probs = c(0.1, 0.5, 0.9))
```

---

qe_curve *Estimate a Quantile–Expectile Calibration Curve*

Description

Estimate the expectile level associated with each empirical left quantile. The calibration curve describes distributional shape and is invariant to translation and positive rescaling.

Usage

```
qe_curve(x, probs = (5:95) / 100, na.rm = FALSE)
```

Arguments

x	A nonempty finite real numeric vector of observations.
probs	A nonempty numeric vector of probabilities strictly between zero and one. Order and duplicates are preserved.
na.rm	A logical scalar. If TRUE, remove NA and NaN before calculation. Infinite values are always rejected.

Details

For the empirical left quantile q_p, let

$$L(p) = n^{-1} \sum_i (q_p - x_i)_+, \qquad U(p) = n^{-1} \sum_i (x_i - q_p)_+.$$

The estimate is $\widehat{\tau}(p) = L(p)/(L(p)+U(p))$. Quantiles use the left inverse convention, corresponding to stats::quantile(type = 1). At sample endpoints the calibration can equal zero or one. For a constant sample the calibration is defined as $1/2$ at every probability.

Ranks are selected by comparing probabilities with represented k/n breakpoints. Calculations use scaled cumulative adjacent spacings for numerical stability. Reported quantiles are on the original data scale.

Value

An object of class `qe_curve`, with components:

probabilities The requested probabilities.

calibration Estimated calibration values.

quantiles Empirical left quantiles.

n Number of observations used.

omitted Number of missing observations removed.

call The matched function call.

The object also retains sample information for exact prediction; internal components are not part of the public interface.

References

Lyu, G. (2026). *Quantile–expectile calibration curves for distributional shape: estimation, inference, and symmetry testing*. Unpublished manuscript, Sections 2–3.

See Also

`qe_asymmetry`, `qe_band`, `qe_methods`

Examples

```
x <- c(-2, -1, 0, 1, 4)
fit <- qe_curve(x, probs = c(0.1, 0.5, 0.9))
fit
as.data.frame(fit)
predict(fit, probs = c(0.25, 0.75))
```

`qe_methods` *Print, Plot, Tabulate, and Predict Calibration Objects*

Description

Standard S3 methods for inspecting, displaying, and evaluating empirical calibration, asymmetry, and confidence-band objects.

Usage

```
## S3 method for class 'qe_curve'
print(x, ...)
## S3 method for class 'qe_asymmetry'
print(x, ...)
## S3 method for class 'qe_band'
print(x, ...)

## S3 method for class 'qe_curve'
plot(x, xlab = NULL, ylab = NULL, main = NULL,
                      col = "#236B8E", lwd = 2, ...)
## S3 method for class 'qe_asymmetry'
```

```
plot(x, xlab = NULL, ylab = NULL, main = NULL,
                          col = "#236B8E", lwd = 2, ...)
## S3 method for class 'qe_band'
plot(x, xlab = NULL, ylab = NULL, main = NULL,
                     col = "#236B8E", lwd = 2,
                     fill = grDevices::adjustcolor(col, alpha.f = 0.18), ...)

## S3 method for class 'qe_curve'
as.data.frame(x, row.names = NULL, optional = FALSE, ...)
## S3 method for class 'qe_asymmetry'
as.data.frame(x, row.names = NULL, optional = FALSE, ...)
## S3 method for class 'qe_band'
as.data.frame(x, row.names = NULL, optional = FALSE, ...)

## S3 method for class 'qe_curve'
predict(object, probs = object$probabilities, ...)
## S3 method for class 'qe_asymmetry'
predict(object, probs = object$probabilities, ...)
## S3 method for class 'qe_band'
predict(object, probs = object$probabilities, ...)
```

Arguments

`x`, `object`	An object of the corresponding class, returned by `qe_curve`, `qe_asymmetry`, or `qe_band`.
`probs`	A nonempty numeric vector of prediction probabilities. Curve probabilities must lie in $(0, 1)$, asymmetry probabilities in $(0, 1/2]$, and band probabilities within the fitted trim interval.
`xlab`, `ylab`, `main`	Axis labels and plot title. `NULL` uses method-specific defaults.
`col`	Curve color.
`lwd`	Curve line width.
`fill`	Fill color for the confidence band.
`row.names`	Optional data-frame row names.
`optional`	Logical; if `FALSE`, enforce valid data-frame column names. The named output columns are valid under either setting.
`...`	For plotting, named arguments passed to the plotting frame, such as `xlim`, `ylim`, or `axes`. Data-frame methods pass these arguments to `data.frame`. Print and prediction methods reject unused arguments.

Details

Prediction recomputes empirical values from retained sample information; it does not interpolate between stored probabilities. Band predictions use the fitted constant half width and clip the endpoints to $[0, 1]$. They cannot extend the band beyond its fitted trim interval.

Plots span the range of probabilities stored in the object. They include the underlying empirical breakpoints, even when those breakpoints were not among the requested probabilities. Asymmetry plots display the open cell values and the possibly different values at the breakpoints.

Value

Print and plot methods return `x` invisibly; plot methods also draw a plot. Data-frame methods return columns `probability`, `quantile`, and `calibration` for curves; `probability` and `asymmetry` for asymmetry objects; and the curve columns plus `lower` and `upper` for bands. Curve and asymmetry predictions return numeric vectors. Band predictions return a data frame with the band columns.

See Also

`qe_curve`, `qe_asymmetry`, `qe_band`

Examples

```
fit <- qe_curve(c(-2, -1, 0, 1, 4))
plot(fit)
as.data.frame(fit)[1:3, ]
predict(fit, probs = c(0.2, 0.8))
```

---

qe_symmetry_test *Test Distributional Symmetry Using Calibration Curves*

Description

Test symmetry using the reflected calibration asymmetry curve and an ordinary empirical bootstrap centered at its observed value.

Usage

```
qe_symmetry_test(x, method = c("cvm", "ks"), trim = 0.05,
                 B = 999L, seed = NULL, na.rm = FALSE)
```

Arguments

`x`	A finite real numeric vector. After missing-value removal it must contain at least two observations and must be nonconstant.
`method`	`"cvm"` for the integrated squared asymmetry statistic (default), or `"ks"` for the absolute supremum statistic.
`trim`	A finite numeric scalar strictly between zero and $1/2$. The asymmetry curve is assessed on `[trim, 0.5]`. Choose this value independently of the data.
`B`	A positive integer number of bootstrap replications.
`seed`	`NULL` or a nonnegative integer seed. With `NULL`, the caller's random-number stream is used and advanced. A supplied seed is scoped to this call, restoring `.Random.seed` and the RNG kind on exit. A cached normal draw from R's `"Box-Muller"` generator cannot be restored; see Details.
`na.rm`	A logical scalar. If `TRUE`, remove `NA` and `NaN`. Infinite values are always rejected.

Details

For the asymmetry estimate $\widehat{D}(p)$ from `qe_asymmetry`, the CvM statistic is

$$n \int_{\text{trim}}^{1/2} \widehat{D}(p)^2 \, dp,$$

with unit weight, and the KS statistic is

$$\sqrt{n} \sup_{p \in [\text{trim}, 1/2]} |\widehat{D}(p)|.$$

The empirical step-function integral and supremum are computed exactly up to floating-point arithmetic. The supremum includes both orientations of the reflected steps. The centered bootstrap process is $W^*(p) = \sqrt{n}\{\widehat{D}^*(p) - \widehat{D}(p)\}$. Its integrated square or absolute supremum supplies the bootstrap statistics. No symmetry center, density, bandwidth, or variance is estimated.

R's `"Box-Muller"` normal generator keeps a cached normal draw outside `.Random.seed`; a scoped seed cannot restore this cache. The default `"Inversion"` normal generator avoids this issue. Use `seed = NULL` when continuation of an existing Box-Muller cache matters. If `.Random.seed` was absent before a seeded call, it is removed on exit. Fixed-seed reproduction also depends on the R random-number settings.

The p-value is $(1 + \sum_b I\{T_b^* \geq T\})/(B + 1)$. Numerical ties count as exceedances. To avoid splitting mathematical ties through rounding, comparison uses a tolerance of $64\epsilon_{\text{machine}}$ on the asymmetry-amplitude scale: $\sqrt{T/[n(1/2 - \text{trim})]}$ for CvM and $T/\sqrt{n}$ for KS. The observed amplitude minus this tolerance is clipped at zero and transformed back to statistic units to give `p.value.threshold`. The reported p-value is exactly:

```
(1 + sum(bootstrap.statistics >= p.value.threshold)) /
  (B + 1)
```

This normalization preserves distinctions on very short trimmed intervals.

This Monte Carlo correction prevents a reported zero but does not provide an exact finite-sample test. It differs from the uncorrected proportions in the manuscript's simulation programs; the application uses the corrected convention. The manuscript treats CvM as the primary test and reports conservative finite-sample behavior for KS.

The theoretical assumptions include iid observations, a finite second moment, a continuously differentiable distribution with positive density on a neighborhood of $[Q(\text{trim}), Q(1 - \text{trim})]$, where Q is the population quantile function, and shape regularity under the null. See `qeshape-package` for the definition of shape regularity. The limiting statistic's distribution must be continuous and strictly cross the critical quantile level. The trim is fixed and deterministic; Monte Carlo consistency requires increasing B. Computational support for ties does not establish these assumptions. With fixed trimming the test detects departures visible on the trimmed interval; nonrejection does not establish global symmetry.

Value

An object of classes `qe_symmetry_test` and `htest`, with components:

statistic The named observed CvM or KS statistic.

parameter A named numeric vector containing `n`, `B`, and `trim`.

p.value The corrected bootstrap p-value.

p.value.threshold The observed-statistic cutoff after allowing for numerical ties, in the same units as `statistic`.

method, alternative, data.name Standard test-description fields.

bootstrap.statistics The `B` centered bootstrap statistics.

B, trim, n, seed The number of replications, trim, number of observations used, and seed.

omitted Number of missing observations removed.

call The matched function call.

References

Lyu, G. (2026). *Quantile–expectile calibration curves for distributional shape: estimation, inference, and symmetry testing*. Unpublished manuscript, Section 4.

See Also

`qe_asymmetry`, `qe_band`

Examples

```
# Small B for a fast example; increase it for an analysis.
x <- c(-3, -2, -1, 0, 0.5, 1, 2, 4)
qe_symmetry_test(x, B = 39, seed = 7)
qe_symmetry_test(x, method = "ks", B = 39, seed = 7)
```

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